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Faculty of Management, Social Sciences and Humanities
Department of Languages
Intake - 41

BSc in Applied Data Science Communication
Applied machine learning (LB 2214)
Year 2 - Semester 4

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Predicting High-Demand Products in E-Commerce Using Customer Purchase Behavior: A Machine Learning Approach

Abstract

As e-commerce becomes more popular, more transactions produce many more data points. As such, there's lots of new opportunities available for better predicting demand and optimising inventory through the use of that transactional data. Despite the opportunities, the task of accurately predicting which products will be the highest demand is challenging for a number of reasons, including sparse data due to not all customers purchasing the same products, imbalanced classes (most products won't sell as much as others), and determining how customers purchase products. This research paper has developed a framework for predicting high-demand products in the e-commerce space by using an aggregated view of all customers who purchase products from e-commerce sites. It uses the Online Retail II dataset, which contains transaction records that were preprocessed to create product-level features that describe the customer's behaviour when purchasing that product. The behaviours include sales volume, number of customers buying the product, number of times the customer purchased the product, and how much revenue the product generated. The revenue generated was then used to classify products into either high- or low-demand categories. The authors used three different supervised machine learning techniques: Logistic Regression, Decision Tree, and Random Forest, and evaluated them based on accuracy, precision, recall, F1-Score, and ROC - AUC score metrics. The experimental results of this study confirm that the Random Forest classification model performed better than the baseline models, achieving a 95.68% accuracy and a ROC-AUC score of 0.993, indicating the Random Forest model has strong discriminative power when dealing with class imbalance. Visualised data also confirms that products with high demand have a very specific behaviour pattern in terms of providing sustained revenue to retailers and having high customer engagement levels. Using the insights gained in this research, we now develop a Smart Goods Recommendation System (SGRS), an interactive data-driven decision support application which allows sellers to identify products that are in high demand by allowing them to leverage contextual filters (e.g. time period, country, expected unit price, top-N selection). As the proposed framework provides a linkage between predictive analytic ability and practical e-commerce-based decision-making, therefore, presents a scalable and easily interpretable solution for demand-aware inventory and promotion strategy development.

I. Introduction

E-commerce platforms are rapidly growing in number, and as a result, the amount of transactional data generated through these online transactions is also rapidly increasing. Transactional data provide insight into how customers buy products and the performance of those products, allowing for improved decision-making regarding inventory management, pricing strategy, and marketing optimization based on data-driven methods. However, the challenge presented by e-commerce transactional data is that they are highly dimensional (highly sparse and dynamic), making it difficult to accurately predict which products will see high demand.

The accuracy of e-commerce demand predictions is essential for effective e-commerce operations. Excessive holding costs and operational inefficiencies are created when too much product is stocked (i.e., high demand products), whereas lost revenue and decreased customer satisfaction are created when too little product is stocked (i.e., low demand products). Many e-commerce sellers still rely primarily on heuristic rules, historical averages, or their own judgement in determining which products to promote or stock, even though there are large amounts of transactional data. These heuristic and historical average methods do not accurately capture the complexities of the non-linear relationships that exist between product performance, customer purchasing behaviour, and contextual variables (such as seasonality, pricing, and regional preferences) that affect demand.

The research challenge that is being addressed in this research relates to the limited use of interpretable and empirically verifiable means for using large-scale transaction data to pinpoint high-demand goods in e-commerce and for translating predictive insight into usable business judgment for e-commerce sellers. The current machine learning approaches to demand forecasting have developed algorithms to accurately predict demand but have not adequately examined the classification of product-level demand, nor the tracking of 'predictive interpretability', nor how to leverage forecasting for a seller-centric decision support system.

Considering current literature, predictive accuracy has been optimized by incorporating demand forecasting and demand-generated networks of behavioural and transactional data to improve prediction accuracy using various supervised and ensemble models. Much of the literature concentrates on improving model performance as an isolated factor and very few provide interactive mechanisms through which sellers can make demand-driven decisions, accounting for contextual restrictions on time period, geographical market, anticipated pricing, and inventory constraints. The research challenge that is being addressed in this research relates to the limited use of interpretable and empirically verifiable means for using large-scale transaction data to pinpoint high-demand goods in e-commerce and for translating predictive insight into usable business judgment for e-commerce sellers. The current machine learning approaches to demand forecasting have developed algorithms to accurately predict demand but have not adequately

examined the classification of product-level demand, nor the tracking of 'predictive interpretability', nor how to leverage forecasting for a seller-centric decision support system.

Considering current literature, predictive accuracy has been optimized by incorporating demand forecasting and demand-generated networks of behavioural and transactional data to improve prediction accuracy using various supervised and ensemble models. Much of the literature concentrates on improving model performance as an isolated factor and very few provide interactive mechanisms through which sellers can make demand-driven decisions, accounting for contextual restrictions on time period, geographical market, anticipated pricing, and inventory constraints.

To tackle the focus of this research, it proposes that machine learning techniques be used to create a framework for the prediction of e-commerce products that will experience high levels of customer demand based on customer behaviour when purchasing products. The framework uses the UCI Online Retail II dataset to conduct thorough data preparation and feature engineering to create useful measures of customer behaviour as it relates to the product level which includes the analysis of purchase frequency, reach to customers, diversity of order, and how much revenue was generated. Based upon the definition of revenue high demand products will be defined as having higher than average levels of revenue and lower than average levels will qualify as being low demand products using a threshold value based upon realistic commercial dynamics present in a typical online retail setting.

Multiple supervised learning algorithms including Logistic Regression, Decision Tree, and Random Forest classifiers were tested for the purpose of evaluating how well they predict high demand products. To allow for the reliable comparison of modelling performance based on class imbalance, the robustness of evaluation metrics used to measure the model's performance will include Accuracy, Precision, Recall, F1 score, ROC AUC. Hyperparameter optimization will be performed to support model generalization and stability.

In addition to predictive modelling, this research has included an application called Smart Goods Recommendation which is an interactive decision support tool for E-commerce sellers that allows for the operationalisation of the prediction of goods that may require demand generations. Users of the Smart Goods Recommendation application can input contextual filters including month, country, predicted unit price, and top N product selection and receive demand probability predictions and, based on the predictions, recommended products.

The integration of predictive insights with accurate predictions will provide for knowledge-based decision making. This integration of predictive insights will also ensure they can be easily visualised, as well as directly implemented into concrete decisions. The contributions of this research are threefold:

1. The development of a product-level model for predicting demand for products based on customer purchase behaviour,
2. The demonstration of the effectiveness of using ensemble-based machine learning models to successfully capture very complex demand patterns through experimentation.
3. The creation of a decision support application that is interactive for sellers and provides a bridge between the academic demand forecasting literature and practical applications of e-commerce.

II. LITERATURE REVIEW

The rapid growth of online shopping platforms has resulted in the generation of massive volumes of transactional data, enabling advanced analytical exploration of customer purchasing behaviour and product demand patterns. The existing literature [1]–[37] spans multiple research domains, including machine learning, deep learning, dynamic pricing, and decision support systems for e-commerce. This section synthesizes these domains to establish a comprehensive foundation for predicting high-demand products in e-commerce environments.

A. Machine Learning Applications in E-Commerce Demand Prediction

Supervised machine learning techniques have been widely adopted for forecasting product demand and sales performance. Ensemble-based models, such as Random Forest, XGBoost, and LightGBM, consistently demonstrate superior predictive performance compared to traditional statistical and regression-based approaches due to their ability to model non-linear relationships and handle high-dimensional feature spaces [1], [4], [7], [10], [31]. Specifically, XGBoost has been successfully applied to purchase forecasting tasks [1], while Random Forest has been identified as a leading classifier when benchmarked against alternative classification techniques [4]. Furthermore, stacking ensemble models that integrate XGBoost, LightGBM, and Random Forest have been shown to outperform individual models by leveraging complementary strengths [7].

Deep learning approaches further enhance forecasting accuracy by capturing sequential and temporal dependencies in purchasing data. Models such as Deep Neural Networks (DNNs) [2], Recurrent Neural Networks (RNNs) [5], [20], Long Short-Term Memory (LSTM) networks [11], [15], [19], Multi-Attention Temporal RNNs (MAT-RNN) [16], attention-based Graph Neural Network (GNN) hybrids [18], and Autoformer architectures [35] have been employed to model

seasonality and long-term demand cycles. Notably, Autoformer reduced RMSE by over 17% compared to LSTM and Informer baselines in cross-border e-commerce forecasting scenarios [35].

Comparative analyses reveal important trade-offs across modeling paradigms. Ensemble methods are generally robust and offer partial interpretability but depend heavily on manual feature engineering [1], [4]. In contrast, deep learning models effectively capture complex temporal patterns but often function as black-box systems [2], [5], [11]. Hybrid approaches attempt to balance predictive power and interpretability, such as integrating LightGBM with SHAP explanations [17] or combining reinforcement learning with LSTM architectures for adaptive demand forecasting [11], [15].

Dataset scale and quality vary considerably across studies. Large-scale datasets, such as those comprising approximately two million transaction records across more than 500 SKUs [35], enable improved generalization but introduce challenges related to data heterogeneity and noise. Conversely, smaller datasets may be susceptible to overfitting and limited external validity [25]. Evaluation metrics also differ across tasks, with regression-based studies emphasizing RMSE, MAE, and R^2 [35], while classification-focused research employs accuracy, precision, recall, and F1-score [4], [10]. This methodological diversity complicates direct cross-study comparisons.

B. Customer Purchase Behavior Modeling

Modeling customer purchase behavior is fundamental to accurate demand prediction. Transaction-level features, including purchase frequency, order quantity, repeat purchase patterns, and customer-product interaction diversity, are commonly utilized [2], [5], [6], [17]. Transaction-driven insights have been examined at both category and product levels [6], while interpretable behavioral feature importance has been extracted using SHAP-enhanced LightGBM models [17].

Sequential modeling approaches emphasize the importance of temporal dynamics in customer decision-making. Reinforcement learning-inspired LSTM models capture adaptive purchasing strategies over time [11], [15], whereas RNN-based frameworks predict future purchasing behavior by leveraging historical sequences [20]. In addition, consumer heterogeneity has been shown to significantly influence demand patterns, with factors such as willingness to wait, geographic proximity, and adoption status playing key roles [27], [33].

Despite these advances, contextual feature integration remains limited. Although some studies incorporate external factors such as promotions, sentiment, or perishability [32], [34], [37], behavioral and contextual variables are often modeled independently. For instance, key drivers of modest fashion demand, including price, retailer, colour, and fabric, have been identified without explicitly modeling temporal seasonality [28]. Similarly, sentiment-aware dynamic pricing

approaches link customer emotions to demand but focus primarily on pricing rather than holistic purchase behavior [32].

Behavior modeling datasets frequently suffer from limited diversity. Many studies rely on single-platform data sources [17], [21], restricting generalizability. Furthermore, the overrepresentation of categories such as fashion and FMCG, coupled with underrepresentation of perishables and durable goods, introduces bias. While accuracy and F1-score are commonly used for classification tasks [4], [10], and RMSE for regression tasks [35], few studies assess fairness, bias, or ethical implications, which are increasingly important considerations in applied AI research.

C. Product Popularity and High-Demand Product Identification

Identifying high-demand products is a central objective in e-commerce analytics. High-demand items are commonly defined using absolute sales thresholds, cumulative revenue contributions, or frequency-based rankings [6], [28]. Classification-based approaches enable products to be grouped into demand categories, facilitating inventory prioritization and targeted marketing strategies.

Machine learning-based classification and forecasting methods generally outperform heuristic ranking techniques. Comparative evaluations demonstrate the effectiveness of GAN, LSTM, and ARIMA models for fashion demand forecasting, with GAN architectures showing superior performance in capturing complex purchasing patterns [19]. Neural autoregressive models have also been shown to outperform Prophet models in short-term forecasting tasks [25]. Similarly, applications of MLP, MQRNN, and Random Forest models to FMCG demand highlight the importance of promotional and calendar-related effects [12].

However, the majority of studies emphasize retrospective analysis rather than real-time or user-driven prediction scenarios [8], [12], [19], [25], [26], [29], [30], [35]. Hybrid optimization frameworks integrating artificial neural networks with fuzzy logic and auction theory have been proposed for cross-border e-commerce contexts [29], but these remain largely theoretical. Collaborative demand risk models for durable goods have also been introduced [30], although empirical validation using transactional data is limited. While dual-channel demand shaping research demonstrates the influence of marketing and service levels on channel choice [33], explicit identification of high-demand products is often absent.

Operational definitions of “high-demand” vary substantially across studies, hindering synthesis. Absolute thresholds and relative performance measures are both employed, yet few approaches integrate behavioral signals such as repeat purchases or sentiment with product-level metrics. This limitation reduces the practical applicability of existing models for anticipating demand surges.

D. Decision Support Systems for E-Commerce Sellers

Several studies investigate decision support systems (DSS) for inventory planning, pricing strategies, and product selection. Surveys and conceptual frameworks have been proposed to guide the integration of analytics into e-commerce decision-making processes [3], [9], [13], [14]. Recommendation systems and demand forecasting architectures emphasize scalability, interpretability, and evaluation challenges [9], [13], while conceptual models support predictive purchase behavior analysis [14].

Dynamic pricing and sentiment-aware systems further contribute to DSS research. Sentiment-aware dynamic pricing frameworks leveraging XGBoost and multilingual sentiment features have been proposed [32], [34], alongside broader reviews of AI-driven pricing strategies and market adaptation mechanisms [36]. The importance of dynamic pricing in agricultural e-commerce, considering perishability and freshness constraints, has also been highlighted [37]. Dual-channel environments demonstrate how marketing and service quality influence demand allocation [33].

Industry applications underscore the practical relevance of these approaches. Dynamic pricing mechanisms are widely employed in platforms such as Amazon, Uber, Netflix, and Airbnb [36]. Empirical evidence from organizational studies indicates that AI adoption improves forecasting accuracy and logistics efficiency [21], while big data-driven forecasting enhances apparel supply chain performance [26].

Despite these advancements, integration of predictive models into interactive, seller-oriented DSS applications remains limited. Few systems incorporate contextual adaptability related to seasonality, regional preferences, or logistics constraints. While SHAP-based interpretability represents progress toward explainable artificial intelligence [17], most deep learning models remain opaque. Although Autoformer demonstrates strong forecasting performance [35], deployment within actionable DSS interfaces remains insufficiently explored.

E. Research Gap Identification

While past experiences have demonstrated the utility of leveraging machine-learning and/or deep learning methodologies in predicting demand and consumer behavior, there are still significant gaps in knowledge about these two research areas. Many studies focus exclusively on enhancing predictive performance and do not put their findings into a guidance decision-making framework through which sellers can derive valuable insights. As a result, these studies have limited practical application in real-world settings. Furthermore, many of the factors that impact demand—such as seasonality; price changes; perishable goods; and differences in demand patterns among

geographic regions-are typically researched individually, which reduces their usefulness in predicting demand in an ever-evolving e-commerce environment.

As mentioned previously, research explicitly focused on determining which products will generate the highest demand in the future based on transaction-level behavioral data is sparse. Additionally, the majority of research that attempts to identify future demand-generating products does not provide the explanation (interpretability) necessary for an informed decision making process. While past research has demonstrated the effectiveness of utilizing classical demand forecasting models to predict future product demand and has also documented the methodology to dynamically price products, developing new methodologies that integrate both dynamic pricing and demand forecasting methodologies into a single prediction will aid in understanding the interaction between supply and demand as a function of all three of these variables.

There has been minimal research to date on the unique challenges associated with cross-border e-commerce. These challenges include logistical delays, exchange rate fluctuations, and heterogeneous consumer behavior across multiple geographic regions.

Finally, although numerous researchers in recent years have begun to utilize large datasets to conduct empirical analysis, there has been little consideration regarding the use of biased or unrepresentative datasets and alternative evaluation metrics (beyond traditional RMSE, MAE and Accuracy) when conducting empirical analysis on demand prediction. Addressing these limitations will enable the development of demand-prediction models that are more robust, equitable, and practical in application.

To address the gaps identified in the literature, this research presents a machine learning-based, interactive decision-making support tool that uses customer's purchase behaviour and contextual factors such as price, geography, and seasonality to assist with forecasting products which will be in high demand. This integrated forecasting tool will create a framework combining behavioural, reputational and pricing dimensions of e-commerce platforms so as to better align forecasted demand on an academic level with operational needs of e-commerce platforms enabling better decision-making around marketing and inventory management.

III. Methodology

A. Introduction to Methodology

The research's aim is to predict which products will have high levels of demand in e-commerce by analysing the customer purchases and providing an evidence-based recommendation for sellers with actionable insights. The data-driven methodology created will accomplish this with the incorporation of data processing, exploration of the data, development of features from the data, the use of machine learning classification (with supervision), and finally, the delivery and interaction with the data via the use of an application that provides decision support. The methodology will meet the research objective by translating the historical data of the purchases into product-level demand predictions, and by making the prediction results accessible through a practical experience for the sellers via an online operational app that informs them on demand.

B. Data Sources and Collection

The study utilizes the **Online Retail II dataset**, a publicly available dataset obtained from Kaggle. The dataset contains historical transactional data from a UK-based online retail business.

- **Data source:** Kaggle (Online Retail II dataset - <https://www.kaggle.com/datasets/mashlyn/online-retail-ii-uci/data>)
- **Time period covered:** December 2009 – December 2011
- **Data type:** Structured transactional data
- **Number of records:**
 - Original dataset: 1,048,575 records
 - After preprocessing: 793,309 records
- **Number of variables:** 8 original attributes, expanded with engineered features
- **Key variables:** Invoice number, stock code, product description, quantity, invoice date, unit price, customer ID, and country

All customer identifiers were anonymized, and no personally identifiable information was accessed, ensuring ethical compliance.

C. Data Preprocessing

Data preprocessing was performed to improve data quality and ensure model reliability:

- **Missing values:** Transactions with missing customer identifiers were removed.
- **Outlier and anomaly handling:** Records with zero or negative quantities and unit prices were excluded to remove returns and invalid transactions.
- **Data type standardization:** Column names were standardized and numerical variables were cast to appropriate numeric types.
- **Feature scaling:** Standardization using *StandardScaler* was applied for Logistic Regression, while tree-based models were trained on unscaled data.
- **Derived attributes:** A revenue feature was created as the product of quantity and unit price.

No text preprocessing was required, as product descriptions were used only for result interpretation.

D. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand demand patterns and feature relationships. Key findings include:

- A **highly skewed revenue distribution**, with a small number of products generating most of the total revenue.
- Strong positive correlations between **total revenue, quantity sold, and unique customers**.
- Clear separation between high-demand and low-demand products based on revenue thresholds.

The following visualization techniques were employed:

- Bar charts (top products, customer revenue)
- Histograms (quantity distribution)
- Scatter plots (revenue vs. quantity)
- Bubble charts (purchase frequency vs. customers)
- Correlation heatmaps

These analyses guided feature selection and model design.

E. Feature Engineering and Selection

Transaction-level data were aggregated at the product level (*StockCode*). The following features were engineered:

- Total quantity sold
- Number of orders
- Average unit price
- Total revenue
- Number of unique customers
- Purchase frequency (quantity per order)

Since modeling occurred at the aggregated product level, categorical variables did not have a direct encoding; Models used no dimensionality reduction techniques due to the inherent low-dimensionality. In this study, the primary approach to feature selection was to use insights from exploratory data analysis (EDA) and to determine which features would be most relevant to the domain of interest, while at the same time preventing future data leakage through the exclusion of total revenue as part of the model's input data.

F. Model Selection

The demand prediction task was formulated as a **binary classification problem**. The following models were evaluated:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

The Random Forest was selected as the final model because it had superior predictive performance and was also more resistant to feature interactions than the other models, therefore it was more suitable to be utilized as the final model.

G. Model Training and Optimization

The dataset was split using **stratified sampling**:

- **Training set:** 80% (3,704 products)
- **Test set:** 20% (926 products)

A 5-fold Cross-validation was used during the training of all models, with hyper parameter tuning being conducted for the Random Forest model through GridSearchCV which eventually optimized tree count, maximum depth and minimum sample split size of said model. 200 trees with no restriction on maximum depth were ultimately utilised for the final optimised model. All models were implemented using the **Scikit-learn** framework.

H. Evaluation Metrics

Model performance was evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

These metrics are appropriate for imbalanced classification problems and provide a comprehensive assessment of both predictive accuracy and class discrimination. The final model achieved **95.68% accuracy** and a **ROC-AUC score of 0.993**, indicating excellent performance.

I. Tools and Experimental Environment

The analysis and model development were conducted using:

- **Programming language:** Python
- **Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- **Development environment:** Local machine
- **Hardware:** CPU-based processing (no GPU acceleration used)

This environment was sufficient given the tabular nature of the dataset and model complexity.

J. Ethical Considerations and Methodological Limitations

The use of publicly available anonymous data allows for adherence to the privacy requirements of the individual and to the privacy requirements imposed on organizations by regulatory bodies and other entities. Limitations may be associated with the model based on the history of the purchasing patterns, and assigning product demand based on the revenue of that product, as well as not specifically considering additional market and promotional shocks to the marketplace.

Although the limitations exist, the current methodology is a reliable and sustainable method to evaluate the stock of products that sell in high volume in the e-commerce space.

IV. Results and Analysis

This portion outlines the experimental findings obtained from evaluating the Online Retail II (UCI) dataset while predicting product demand trends through customer purchase history. Various machine learning algorithms are benchmarked, and a rigorous examination of graphical evidence that validates the effectiveness of the proposed Smart Goods Recommendation System is conducted.

A. Dataset Characteristics After Preprocessing

The complete dataset comprised of 1,048,575 purchase records prior to any pre-processing event; however, following the elimination of records containing missing customer IDs, incorrect units sold, and irregular price points, 793,309 transactions were left for further analysis. These transactions were subsequently aggregated at the item level to produce 4630 distinct items. Items were represented by a series of engineered attributes which were created from customer purchasing histories, such as: the total number of items purchased; the number of purchase orders; average item price; total revenue generated for each item; the number of unique buyers; and the frequency in which an item is purchased. Products were labeled as *high-demand* if their total revenue exceeded the **75th percentile threshold**, resulting in:

- **1,158 high-demand products (25%)**
- **3,472 low-demand products (75%)**

This skewed distribution reflects realistic e-commerce environments, where a limited number of products account for a large proportion of total sales.

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\dulmi\Downloads\online shopping>
python -u "c:\Users\dulmi\Downloads\online shopping\tempCodeRunnerFile.py"
Dataset Shape: (1048575, 8)
Invoice StockCode Description Quantity InvoiceDate Price Customer ID Country
0 489434 85048 15CM CHRISTMAS GLASS BALL 20 LIGHTS 12 12/1/2009 7:45 6.95 13085.0 United Kingdom
1 489434 79323P PINK CHERRY LIGHTS 12 12/1/2009 7:45 6.75 13085.0 United Kingdom
2 489434 79323W WHITE CHERRY LIGHTS 12 12/1/2009 7:45 6.75 13085.0 United Kingdom
3 489434 22041 RECORD FRAME 7" SINGLE SIZE 48 12/1/2009 7:45 2.10 13085.0 United Kingdom
4 489434 21232 STRAWBERRY CERAMIC TRINKET BOX 24 12/1/2009 7:45 1.25 13085.0 United Kingdom
After Cleaning Shape: (793309, 9)
StockCode Total_Quantity_Sold Number_of_Orders Avg_Unit_Price Total_Revenue Unique_Customers Purchase_Frequency
0 10002 8624 297 0.841388 6761.52 164 29.037037
1 10080 303 26 0.509259 124.61 23 11.653846
2 10109 4 1 0.420000 1.68 1 4.000000
3 10120 658 61 0.210000 138.18 52 10.786885
4 10123C 632 46 0.624000 229.36 39 13.739130
High_Demand
0 3472
1 1158
Name: count, dtype: int64

```

B. Model Performance Evaluation

Three supervised learning algorithms were compared for predicting high demand on products - Logistic Regression, Decision Tree and Random Forest, using Accuracy, ROC-AUC, Precision, Recall and F1-score as measures of model performance.

1) Logistic Regression Model's accuracy was 91% with a ROC-AUC of 0.96. While it did provide a good baseline, recall for high demand products (i.e., 0.77) was lower compared to models based only on Random Trees or Decision Trees. As such, it is clear that there will be high value items that are misclassified as low value.

Logistic Regression					
Accuracy: 0.9103671706263499					
ROC AUC: 0.9603622180264335					
	precision	recall	f1-score	support	
0	0.93	0.96	0.94	694	
1	0.86	0.77	0.81	232	
accuracy			0.91	926	
macro avg	0.89	0.86	0.88	926	
weighted avg	0.91	0.91	0.91	926	

2) Decision Tree Classifier - With higher classifying capabilities than a Logistic Regression model, accuracy was 95% and AUC was 0.93. Overall, recall for high demand products (i.e., 0.88) was improved but there is a potential risk of overfitting due to the structure of the Decision Tree classifiers.

Decision Tree				
Accuracy: 0.9524838012958964				
ROC AUC: 0.9295625062108716				
	precision	recall	f1-score	support
0	0.96	0.98	0.97	694
1	0.92	0.88	0.90	232
accuracy			0.95	926
macro avg	0.94	0.93	0.94	926
weighted avg	0.95	0.95	0.95	926

3) Random Forest Classifier- With the greatest performance, achieving 95.9% accuracy and AUC 0.993. Creating strong balance of precision (i.e., 0.95) and recall(i.e., 0.88) for high demand products demonstrates a robust ability to generalize while capturing the complexities of non-linear purchasing behaviour tendencies of customers.

Random Forest				
Accuracy: 0.958963282937365				
ROC AUC: 0.9934444251217331				
	precision	recall	f1-score	support
0	0.96	0.98	0.97	694
1	0.95	0.88	0.92	232
accuracy			0.96	926
macro avg	0.96	0.93	0.94	926
weighted avg	0.96	0.96	0.96	926

C. Hyperparameter Optimization and Final Model Selection

Hyperparameter tuning using GridSearchCV further optimized the Random Forest classifier. The optimal configuration consisted of:

- 200 estimators
- No maximum depth restriction
- Minimum samples split of 5

The optimized model achieved a final **accuracy of 95.7%** and a **ROC–AUC of 0.993** on the test dataset. The confusion matrix analysis confirmed high classification accuracy for both demand classes, validating the Random Forest model as the final predictive engine for the proposed system.

```
Best Parameters: {'max_depth': None, 'min_samples_split': 5, 'n_estimators': 200}
```

```
FINAL MODEL PERFORMANCE
```

```
Accuracy: 0.9568034557235421
```

```
ROC AUC: 0.9931742522110703
```

	precision	recall	f1-score	support
0	0.96	0.99	0.97	694
1	0.95	0.87	0.91	232
accuracy			0.96	926
macro avg	0.96	0.93	0.94	926
weighted avg	0.96	0.96	0.96	926

D. High-Demand Product Identification

Using the optimized Random Forest model, demand predictions were generated for all products. The model successfully identified several high-revenue products as high-demand, including:

- *REGENCY CAKESTAND 3 TIER* (£282,511.72)
- *WHITE HANGING HEART T-LIGHT HOLDER* (£250,513.22)
- *JUMBO BAG RED WHITE SPOTTY* (£169,776.24)

These products exhibited high purchase frequency, large customer reach, and consistent revenue generation, demonstrating the model's effectiveness in identifying products suitable for promotion.

E. Low-Demand Product Detection

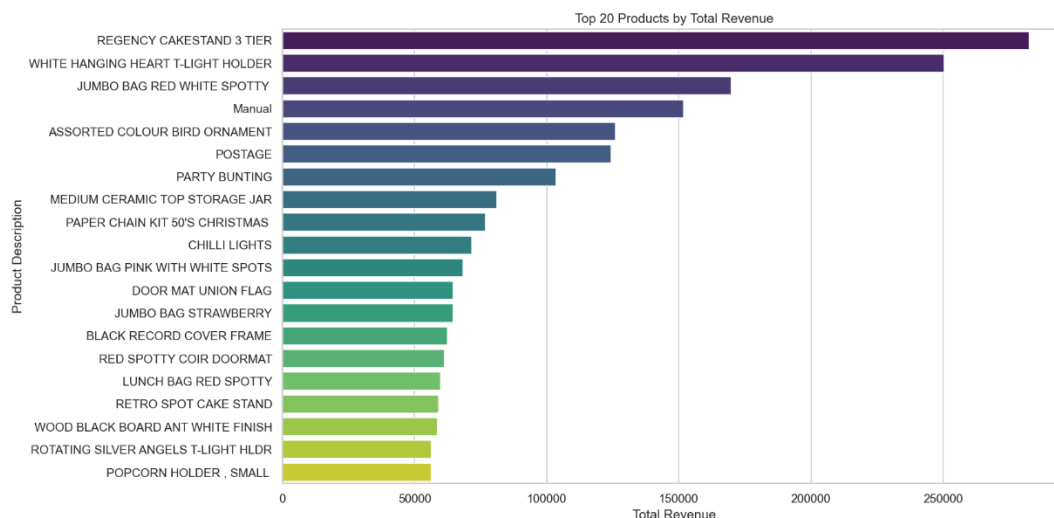
The low revenue generating products that were included in the model examples often produced sales < £ 2. Examples of these include seasonal items, test products and niche accessories (PAD'S TO MATCH ALL CUSHIONS, BENDY COLOUR PENCILS). The results of this analysis provide the opportunity for retailers to take action on inventory clearance, discount strategy and or the decision to stop stocking.

F. Visual Analysis and Insights

The exploration of visual data was an important part of validating the results of the machine learning process as well as identifying the characteristics of customer purchasing behaviour that create demand for a product. The visualised data plots corroborate the predictive capability of the proposed system as well as providing a basis for decision-making in business by giving an interpretation of the information derived from the data.

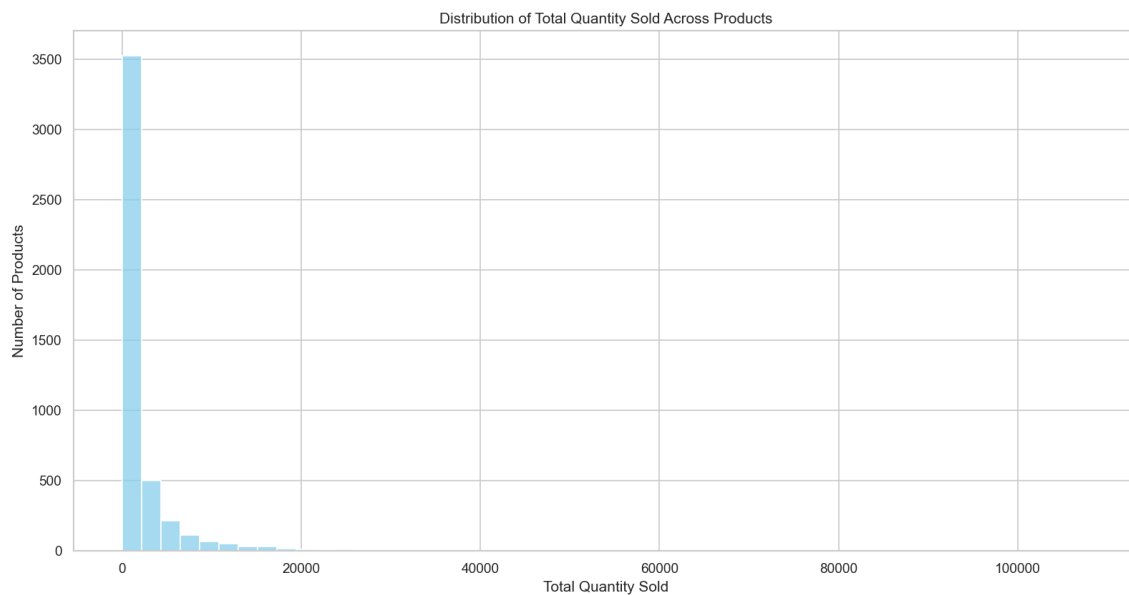
1) Revenue Concentration Among Products

The bar graph that depicts the total revenue generated by the top 20 products should demonstrate a significant concentration of revenue across a small number of products, which accounts for a large percentage of total sales. The REGAL CAKESTAND 3 TIER and WHITE HANGING HEART T-LIGHT HOLDER generated notably higher revenues than other products. This observation supports the commonly referred to principle of the Pareto Principle which is a proven fact within retail—approximately 20% of products generate the majority of revenue. Therefore, this visualisation suggests that these products should be prioritised when developing a promotional strategy and/or featured listing within the Smart Goods Recommendation App.



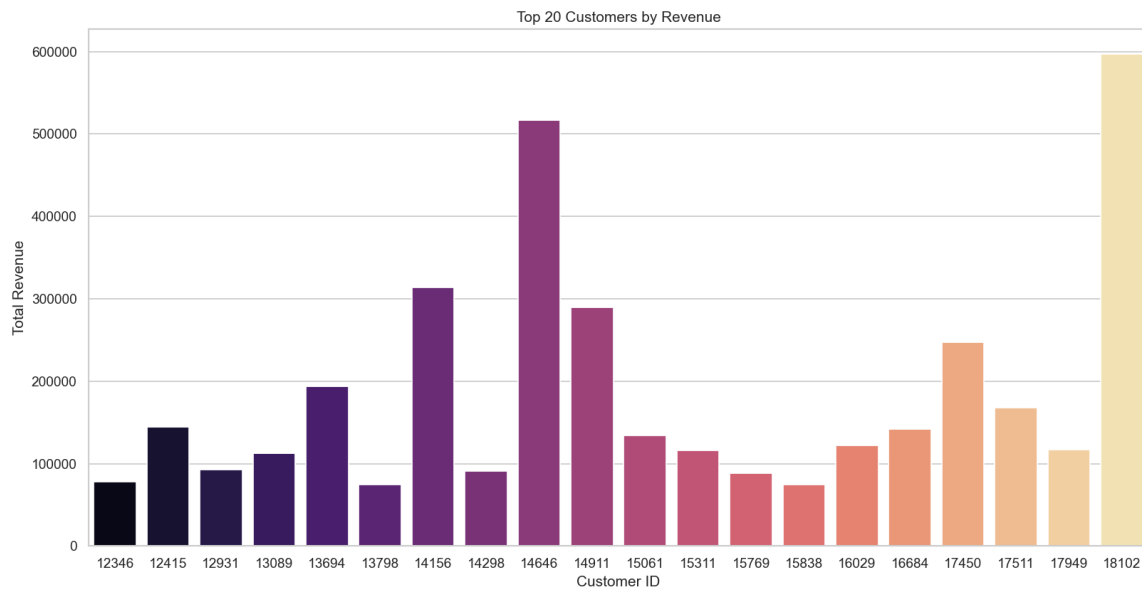
2) Distribution of Sales Volume Across Products

The histogram for the numbers sold depicts a significant amount of right skewness. Most products have relatively low sales volumes, whereas a small number of products exhibit relatively large sales volumes. This presents an imbalance in total sales volume per product and presents challenges when trying to anticipate demand for these products in the e-commerce market since most products contain few historical indicators regarding demand and behaviour. As indicated in the diagrams above, engineered behavioural attributes and ensemble-based models will work best when working with scarce and highly skewed data.



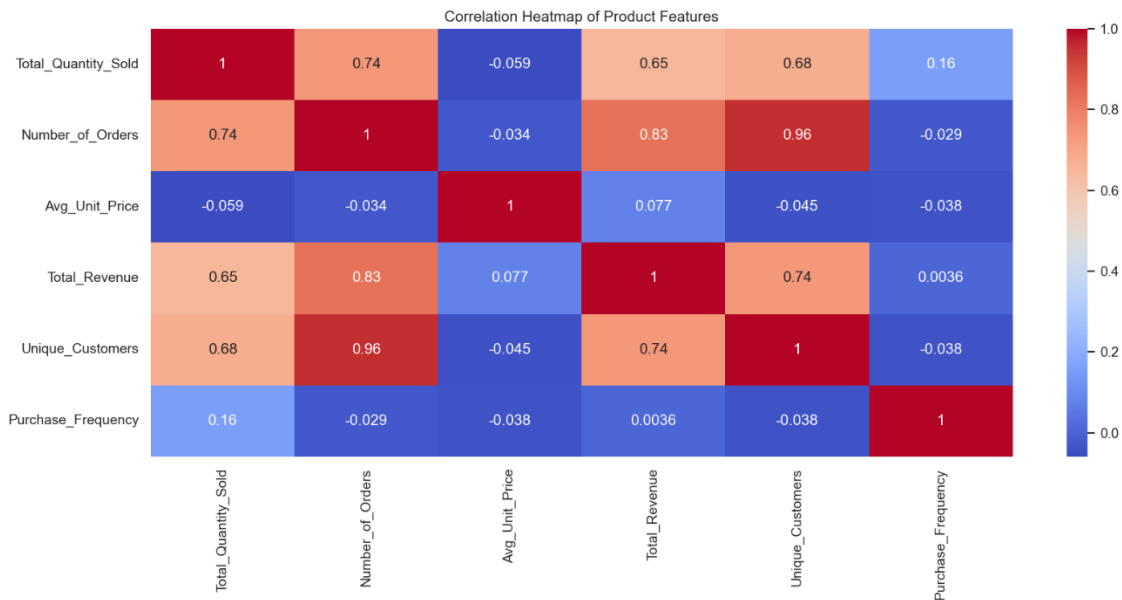
3) Customer Revenue Contribution Analysis

Using a graph based on data to show the 20 customers who have generated the most total revenue shows us that there is a limited number of customers who are responsible for the majority of the sales. The graph also supports the idea that, while customer purchasing behavior is heterogeneous (i.e., each customer is different), it is not uniform (i.e., all customers behave alike). Therefore, an important finding related to customer purchasing behavior is that a small group of repeat customers is necessary to maintain product demand in the long term. While this study does not directly address customer-level personalization strategies, the data suggest that an extension of the current system to include customer segmentation and personalized recommendation strategies is likely to occur in the future.



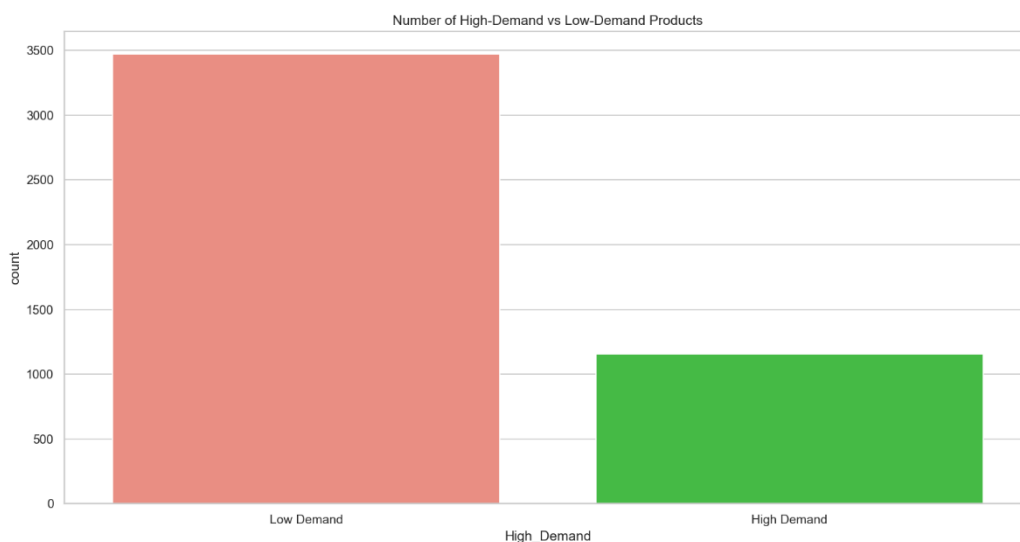
4) Feature Relationship and Correlation Analysis

The correlation heatmap represents the relationship between the features at the product level via quantifiable numerical values. The correlation heatmap shows a very strong positive correlation between total quantity sold and total revenue, supporting the conclusion that the quantity of products sold is the major driver of profit. The moderate level of correlation found between unique customers, number of orders, and purchase frequency indicate that both the ability to reach a large number of customers and their likelihood of returning to purchase are influential factors on the demand for a product. In addition, there are no significant multicollinearity issues associated with the product-level features selected; therefore, the product features selected for inclusion in a machine learning model have been deemed appropriate, which provides confidence in the predictive capabilities of the machine learning model.



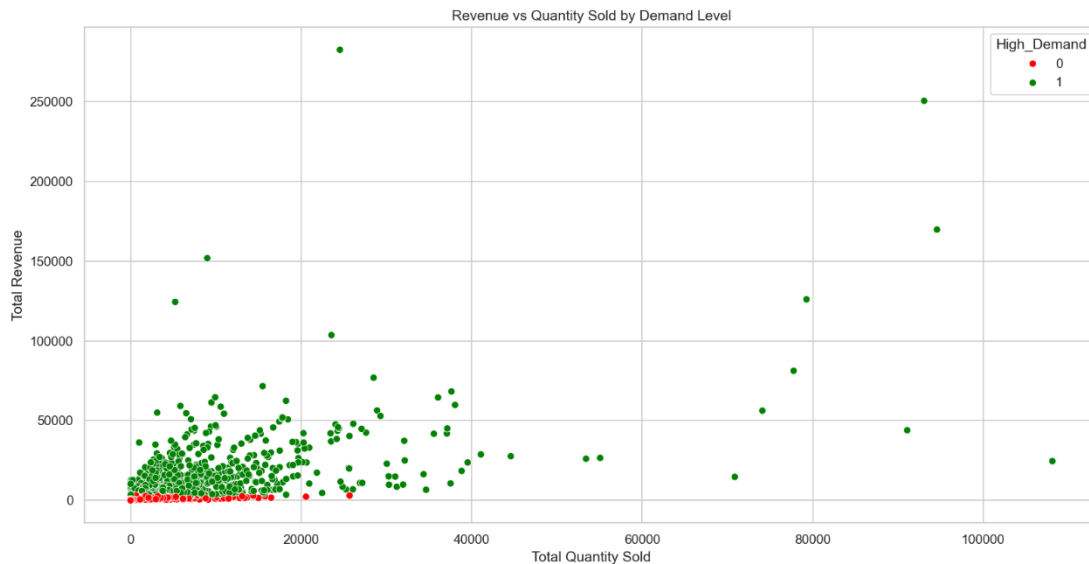
5) Demand Class Distribution

In the count plot that compares high demand and low demand labelled products, one can see a natural imbalance between the high demand and low demand classes, as the high demand category accounts for only 25% of all product classes. The plot indicates that the high demand category is relatively rare, but also shows how commercially significant the earth is. This imbalance makes it necessary to measure the performance of classification models using metrics such as ROC–AUC and F1-score; therefore, these measures are more relevant than using accuracy.



6) Revenue–Quantity Relationship by Demand Level

The relationship between total revenue and total quantity sold is presented in a scatter plot, coloured according to demand class. The scatter plot allows for an easy separation between high-demand and low-demand labelled products. The demand products cluster in higher revenue and higher quantity regions, whereas the lower demand products are located in lower price/less revenue regions. The separation validates the labelling methodology and indicates that the predictions made by the model are in accordance with economic trends as represented in the data.

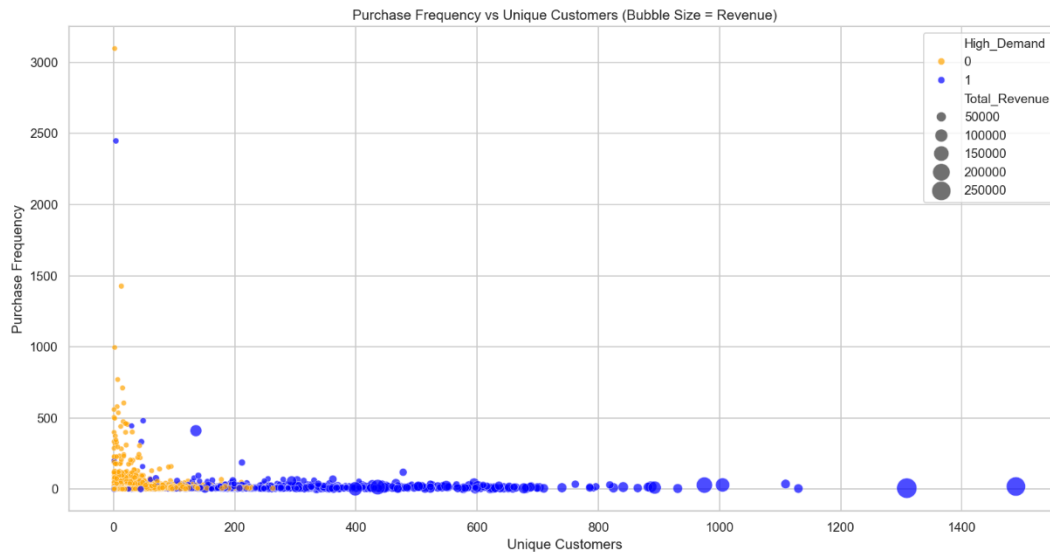


7) Customer Engagement and Purchase Frequency Analysis

Also displayed is a bubble chart that plots the number of unique customers by purchase frequency; bubble size indicates how much revenue each cross represents (from \$bil A). This is the first of many charts that give a three-dimensional view of the performance of products sold through the app. High demand products tend to have:

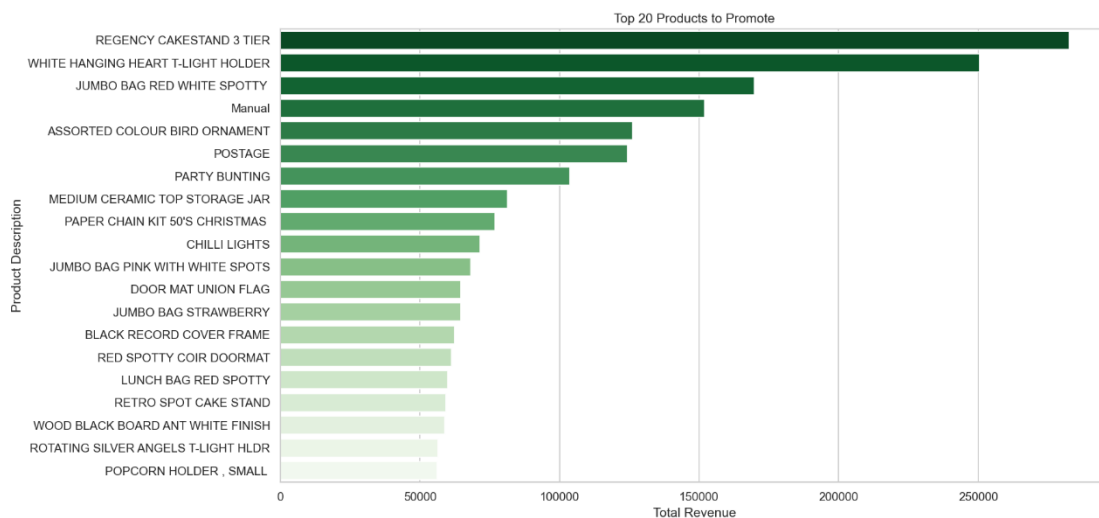
- A large number of customers
- A high frequency of purchases
- A Larger Contribution in Revenue

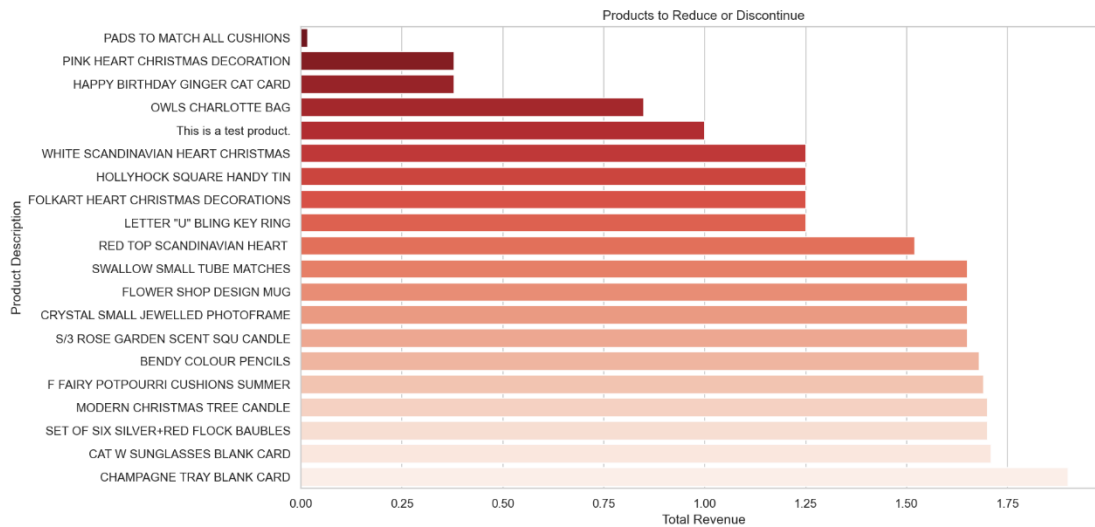
As these charts illustrate, repeat purchases made by customers and positive engagement levels drive sustained demand for these types of products. The insights provided by this type of visualizations helped inform the decision on which behavioral features to use in predictive modeling.



8) Promotion and Discontinuation Insights

The bar graph provides an overall view of the complete product assortment available to a seller, along with suggested promotions and suggested items to discontinue. Products that are in high demand continue to produce strong revenues historically, while items that do not have a high level of demand generate minimal revenue and take up space in a seller's inventory. Using these types of visualizations allow sellers to align their marketing and promotional activities on high-value items as they see where to strategically place marketing efforts. Using these types of visualizations allows sellers to better allocate their inventory to be able to make use of all distributable items; to support the operational goals of the Smart Goods Recommendation app.





9) Alignment with Application Workflow

The collective visualizations presented, assist in reinforcing the overall design assumptions used in creating the application workflow. (See demand probability estimates, Top-N Product recommendations and Product Rankings). All of these items are based on visible patterns from the visual analysis. This ensures that the system remains transparent, interpretable, and practically relevant for e-commerce sellers.

G. Implications for the Smart Goods Recommendation System

The results confirm that customer purchase behavior can be effectively leveraged to predict product demand in e-commerce environments. The proposed system enables sellers to:

- Estimate demand probability for products
- Identify top-N products for promotion
- Optimize inventory management through data-driven insights

By integrating predictive modeling with interactive filtering options such as time period, country, expected price, and top-N selection, the *Smart Goods Recommendation* app translates complex analytical results into actionable business decisions.



Smart Goods Recommendation

Upload Dataset

No file chosen

Upload & Train

Dataset uploaded & model trained successfully!

Get Recommendations

Month

2

Expected Price

3

Country

Australia

Top N Products

10

Get Recommendations

Get Recommendations

Demand Probability (%)

Product	Probability %	Revenue
12 PENCIL SMALL TUBE WOODLAND	100.0	211.2
BAKING SET SPACEBOY DESIGN	100.0	306.0
BATHROOM METAL SIGN	100.0	6.6
BEWARE OF THE CAT METAL SIGN	100.0	20.28
BUTTON BOX	100.0	195.0

Top 10 Products

Product	Revenue
REGENCY CAKESTAND 3 TIER	282511.72
WHITE HANGING HEART T-LIGHT HOLDER	250513.22
Manual	151913.11
JUMBO BAG RED RETROSPOT	136139.64
ASSORTED COLOUR BIRD ORNAMENT	126052.47
POSTAGE	124442.04
PARTY BUNTING	103627.78
MEDIUM CERAMIC TOP STORAGE JAR	81243.22
PAPER CHAIN KIT 50'S CHRISTMAS	76906.96
CHILLI LIGHTS	71631.82

V. Discussion

The aim of this study was to determine if customer purchase behaviors will provide insight into forecasting demand for e-commerce products, as well as to assess different types of machine learning technologies to identify products that will be in high demand and will assist businesses by making inventory and promotional decisions based on data-driven methods. The results indicate that examining transactional purchase behaviors to classify product demand is achievable and will provide businesses with additional opportunities.

A. Research Objective Versus Study Findings

The Primary Objective was to forecast demand for an e-commerce product or service using the historical purchase behaviors of those customers who purchased the same product and then to utilize those forecasts as decision-support applications. The study revealed that engineered behavioral attributes such as total revenue, total sales volume, total customer reach, and frequency of purchases could be used to create a statistical measure for distinguishing between high-demand products and low-demand products, resulting in highly accurate forecasts of demand. These forecasts demonstrate that customer purchase behaviors can be used to accurately represent product demand in e-commerce environments.

Moreover, the findings that ensemble-based" prediction models outperformed the other prediction models in terms of likelihood of being able to forecast demand patterns are consistent with the hypothesis that demand patterns related to retail are likely to be non-linear and heterogeneous in nature. Therefore, these forecasts may also be used successfully by sellers within an interactive application environment as a means of converting analytical findings to actionable recommendations.

B. Interpretation of Key Outcomes

Among the various models assessed, Random Forest classification exhibited the highest degree of stability across all metrics. This indicates that ensemble Learning Strategies have an inherent advantage with respect to undertaking demand forecasts for products with multiple features and imbalanced classes. Previous eCommerce Demand Forecast Analysis Studies have documented

similar findings when looking at ensemble models using trees. Both demonstrate their robustness when forecasting using highly-skewed and/or noisy transactional data.

When looking more closely at the Revenue-Centric features and the Demand Classification of products, it can be seen that not only does a high demand product have a high sales volume, but also a continuous level of customer engagement. Moreover, the presence of moderate-to-strong correlations among Customer-Centric features indicates that Customer Repeat Purchase Behaviour and Customer Diversity are contributing factors that play a key role in the development of Demand for products. The results from this study are consistent with findings from the research literature about the significance of Behavioural Aggregation when undertaking Retail Analytics.

Surprisingly, products with lower relative sales volume, but higher levels of customer reach (i.e., a higher number of Unique Consumers), have also been classified as having a higher Demand Classification, indicating that a product's Demand Classification is not exclusively tied to the amount of product sold. This further emphasises that there is significant value in collecting and utilizing multiple Behavioural Dimensions rather than relying solely upon sales counts in order to develop the most effective Demand Forecasting/Estimation Model.

C. Significance and Practical Implications

The results of this study indicate that the application of predictive demand modeling in an online retail setting allows for superior inventory and marketing decisions via eCommerce. The identification of a limited subset of highly impactful products enables retailers to focus on marketing and promotes optimized inventory levels while also lowering their cost of inventory holding. Furthermore, retailers can look strategically at how to manage consistently low demand items through discounting, bundling, or discontinuation based on their analysis of the company's predictive output.

By implementing the predictive output from the model into an interactive user application with a simple, easy-to-navigate interface, the predictive model will allow retailers who do not have a background in the technical field to access and utilize the model output, as well as to use the tool for making real-time exploratory decisions based on immediate customer needs. The intention of the system is to provide supplementary, supportive guidance for retail decision-making through clear, comprehensive, and documented recommendations, rather than replacing human rationale or instincts.

On a theoretical level, this research contributes to the expanding body of literature concerning behavioral data-mining in eCommerce by demonstrating that demand can be effectively modeled at the product level without requiring customer-level personalization or sophisticated temporal prediction methods.

D. Limitations of the Study

The limitations of this analysis should be kept in mind when interpreting the findings of the study. First, because it is based on historical transactional data, the analysis does not consider how future demand may shift due to external forces (such as seasonality, market impact, or promotional campaigns). Therefore, the model predictions should be viewed as probabilities rather than certainties.

The second limitation is that the data set has a natural class imbalance because of the low percentage of items that experience high demand. While this represents the actual situation of a retail environment, it could create sensitivity issues for the model towards particular product categories. Furthermore, the model's inability to incorporate explicitly time-related features suggests that it will not be able to accurately account for short-term fluctuations in demand.

Thus, although the proposed methodology has some utility in supporting strategic decisions, it will require additional refinement to make it appropriate for predicting short-term fluctuations in demand.

E. Ethical and Responsible Use Considerations

Researchers conducting this study utilized a de-identified dataset, which did not contain any personal information about any of the customers included in the dataset; therefore, researchers felt comfortable with the data they were using and minimised any potential privacy concerns associated with it. While the concentration of revenue across a small number of customers and products does raise some ethical issues relating to fairness of treatment among customers and products, researchers also recognise that over-reliance upon demand forecasting may result in unintentional disadvantages for products that are classified as "niche" or "emerging", if the forecast is made without appropriate context of the product's or the seller's situation at that time. When utilising predictive recommendations, researchers also believe that these tools should be viewed as tools to support decision making, rather than to act as a replacement for decision making. By providing transparency and allowing users to create their own filtering criteria, such as those

present in the proposed application, researchers believe that ethical risks associated with over-reliance on predictive recommendations will be minimized because human oversight will be maintained.

F. Directions for Future Research

There are several avenues of research that could continue from the current work to refine the findings of this paper. Future research could improve upon the current results by incorporating temporal modelling techniques into the research and thereby enabling the estimation of seasonal demand patterns and/or evolving demand over time. Incorporation of time-series features or recurrent model techniques could improve responsiveness to the shorter-term demand fluctuations in the marketplace. Future research could also be enhanced by combining customer segmentation with product level demand estimation in order to develop more targeted and personalised recommendation models. Future research could also include either reproducing the model or testing the model across multiple datasets in various retail or market environments to determine generalisability of the recommendations made by the created model, as well as by including non-time factor variables (e.g., price changes, promotional offers, and supply chain disruption situations) to provide a real-time context for how this system could be applied in a commercial business.

VI. Conclusion

Using customer purchase behavior, this study proposed a machine-learning system to help retailers forecast customer interest in eCommerce products. Research has shown that by analysing historical transaction data (via the machine learning model) along with behavioral characteristics, it is possible to generate actionable intelligence for sellers about how to create or stock products.

The research compared different supervised learning techniques to build predictive models of product demand in eCommerce environments. Random forest was found to be the best-performing technique across all of the analytical metrics used to evaluate models. The study also demonstrates that using ensemble methods to capture the complexities of customer purchasing behaviour is highly beneficial for forecasting product demand, particularly when there are imbalanced classes and heterogeneous customer demand patterns in an eCommerce dataset.

Beyond just predicting product demand, this research built an interactive decision-support tool for sellers of eCommerce products that allows sellers to filter by period, country, projected price, and top-N products to obtain recommendations for stocking high-demand items. By allowing sellers to identify both high- and low-demand products from the same analysis, inventory management decisions, targeted promotional efforts, and cost optimisation can be made with confidence.

The study's results support the prospect of using behavioral data to improve demand forecasting. However, the study's limitation includes the heavy dependence on historical data and lack of definite consideration of time in the modelling. Future use of time-sensitive information, elements of customer segmentation, and consideration of external events, will likely yield increased predictive accuracy and enhanced flexibility.

This research clearly shows that customer purchase patterns and behaviours serve as a reliable basis for developing eCommerce demand responsive recommendation systems. This framework provides both the academic community and the retail industry with the ability to conduct ethical decision making and to have insight into how to apply data driven decision making at scale, interpretability, and tort law compliance.

VII. Limitations and Future Work

While the results of this research are very promising there are a number of limitations that need to be addressed.

The first limitation of the suggested approach is that it solely utilizes historical transactional data as the basis for predicting product demand. As a result, the suggested method restricts solely to observable customer purchase behaviour; it does not consider any extrinsic factors, such as unanticipated changes in market conditions, promotional actions by competitors, disruptions in the supply chain, or shifts in customer preferences that may materially affect product demand patterns over time.

The second limitation is based on how demand is labeled. The approach uses a threshold to classify revenue generated by product sales based on revenue distribution from the dataset used in this study. Thus, the method employed to classify high demand products, while a representation of actual retail activity may not generalize to other markets, product categories or business models. Many low revenue contributing products, because they are strategically important to the retailer, may be misclassified as being of low demand and therefore not categorized in the high demand

product class. Finally, the class imbalance inherent to the dataset, although it is a representative sample of many online retailers in the real-world setting may introduce a bias with regard to the model's sensitivity to low demand classes. Also, the models have been trained and tested on a single dataset, which restricts the ability to generalise the findings across multiple platforms, regions or across borders for those that do business internationally.

From a technical standpoint, while Random Forest and other ensemble models, are able to produce excellent predictive performance and are thus quite robust, on the other hand, they require greater computational resources than their simpler counterparts, thus they are considerably more expensive to operate at scale and require significant organisational investment to implement successfully (i.e., iteratively adjust the model parameters until optimal; i.e., HyperParameter Tuning). And while feature importance can still be generated through ensemble modelling, their interpretability is still much more constrained than it is for linear modelling approaches.

Future research studies can address these limitations by introducing temporal-based modelling approaches to account for seasonality and changes in demand trends over time. Employing time series data features or developing hybridisation of machine learning and deep learning applied to time series forecasting will improve forecast accuracy over the short term. Additionally, extending the existing framework to allow for customer segmentation or for developing personalised recommendations based upon customers may increase relevancy and adaptability.

Finally, additional future research may test the proposed approach through evaluation of multiple datasets, product verticals, or geographical regions to assess its generalisability and robustness. The use of external contextual factors including Promotional Activity, Dynamic Pricing, and Logistics Constraints in addition to sentiment indicators will provide more insight into the dynamics of demand than using only internal data. The incorporation of fairness-aware evaluation metrics and Bias Analysis provides for a more ethical and equitable approach when deploying demand prediction systems in the E-Commerce environment.

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